Project TIWALA: A Data-Driven, Socio-Technical Framework for Agricultural Intelligence in Isabela, Philippines

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***Abstract*—Smallholder farmers in Isabela, Philippines, face significant economic and climate vulnerability, despite high regional productivity. A critical "trust and usability divide," not a lack of hardware, hinders technology adoption among a skeptical demographic (average age 58) that relies on peer-to-peer knowledge. This paper introduces Project TIWALA, a data-driven, socio-technical framework designed to bridge this gap by engineering trust through its core software architecture. We propose a multi-layered system: (1) A Dynamic Farmer Profile Engine uses K-Means clustering on survey data to create statistically-derived user personas (e.g., "The Traditionalist," "The Digital Adopter"), enabling hyper-personalization. (2) A "Safety by Design" (SbD) advisory engine, built on Retrieval-Augmented Generation (RAG), synthesizes information from a curated knowledge base of verified government and research sources (e.g., DA, PAGASA). This RAG layer is the core trust mechanism, grounding Generative AI to prevent misinformation and address farmer concerns about "unsure if true" data. (3) A "Bayanihan" community feature digitizes the existing, trusted peer-to-peer network, leveraging social influence. (4) A multi-channel delivery layer dynamically serves advisories via a rich app or simple SMS, matching the farmer's data-driven persona and digital literacy. By designing these integrated, data-driven layers to directly address the root causes of skepticism, TIWALA provides a replicable model for fostering technology adoption by establishing verifiable trust.**

Keywords—Agriculture Intelligence, Climate Resilience, Decision Support Systems, AI, Human-Computer Interaction, Socio-technical Systems, Technology Adoption, Trust, Retrieval-Augmented Generation (RAG), Data-Driven Personas, Isabela, Philippines

I. Introduction

The province of Isabela is an integral part of the Philippines' agricultural sector and national food security [1]. Located in the Cagayan Valley, it is the country's top producer of corn and second-largest producer of rice [1]. This output has significant economic value; Isabela ranked second nationwide in the value of crop production, generating PhP 59.50 billion in 2023 and increasing to PhP 61.73 billion in 2024 [3].

At the regional level, Isabela accounted for 50.1% of the Cagayan Valley's total value in Agriculture, Forestry, and Fishing in 2022 [5]. This is supported by a large agricultural infrastructure, with the province having the most farms (129.7 thousand) and the largest area of agricultural land (240.6 thousand hectares) in the region as of 2002 [6]. With a rice sufficiency rate of 224%, Isabela produces more than double what its population consumes, supplying essential rice to Metro Manila and other provinces [1].

Despite the province's high production levels, the individual farmers who generate this wealth face significant economic strain. The average net return for rice farming in Isabela is only PhP 15,982 per hectare per season, which is less than half the national average of PhP 36,211 [7]. This is worsened by low farmgate prices for palay (unmilled rice). In August 2025, the price in Isabela was PhP 14.27 per kilogram, well below the national average of PhP 17.11 [7].

National policy, particularly Republic Act No. 11203 (the Rice Tariffication Law), has contributed to this problem by liberalizing rice importation. This has led to an influx of cheaper imported rice, forcing local farmers to lower their prices to compete [8]. When surveyed, farmers in Isabela identified their biggest challenge as the "High Cost of Inputs & Low Price of Produce," directly linking national policy to their financial difficulties [2]. The province's focus on rice and corn has made its over 113,000 rice farmers particularly vulnerable to market shocks affecting these specific commodities [2].

Table 1 summarizes the key indicators that illustrate this contrast between high provincial output and low individual farmer returns.

Metric	Value
Value of Crop Production (2024)	PhP 61.73 billion
National Rank in Crop Production	2nd
Share of Regional Agri-GDP (2022)	50.1%
Total Palay Production (Cagayan Valley, 2024)	2,922,717 metric tons
Number of Rice Farmers	~113,000
Average Farm Size	1.86 hectares

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National Average Net Return (per ha/season)	Php 36,211
Isabela Average Net Return (per ha/season)	Php 15,982

Table 1: Agricultural and Economic Profile of Isabela Province (2022-2024) [Table Placeholder]

In addition to economic pressures, farmers in Isabela face increasing threats from climate change. A "Climate Risk Vulnerability Assessment of the Agriculture Sector in Isabela Province" identified high exposure to typhoons, floods, and erosion [11]. Farmers in the region have long dealt with both flooding and drought, and climate change has made these weather events more frequent and severe [12]. The province is located in the Cagayan River Basin, which is projected to experience an increased risk of both dry and wet spells, worsening the cycle of drought and flooding [14]. Local communities have observed these changes firsthand, noting more frequent droughts and powerful typhoons that cause destructive floods [11].

In response, the government has implemented large-scale infrastructure projects. The Pasa Dam Small Reservoir Irrigation Project (Pasa SRIP), a PhP 1.029-billion project completed in 2018, is a key example [16]. Funded mainly by the Korea International Cooperation Agency (KOICA), the project was designed as a climate change adaptation strategy to manage water resources [12]. The dam aims to mitigate flooding and provide reliable irrigation for 846 hectares, benefiting 619 farmers and increasing annual rice production [16].

However, these solutions, which focus on physical infrastructure, address water availability but do not resolve the imbalance of information at the farm level. The full benefit of these investments is only realized when farmers have the necessary information to use these resources effectively. The presence of a dam, for instance, does not guarantee its optimal utilization. This gap is highlighted by a surveyed farmer who, despite having access to irrigation, asked, "Kailan ang patubig?" ("When is the irrigation schedule?") [2]. This shows a clear need for a complementary IT application layer—a system to deliver timely, actionable information that turns large-scale infrastructure into farm-level benefits.

The problem in Isabela is not a lack of access to technology. Survey data shows that 96% of farmers own a smartphone, despite an average age of 58 [2]. This finding challenges the common assumption of a hardware-centric perceived requirement. The real barriers are related to trust and usability.

Educational attainment is relatively low, with 52% having finished high school and 24% only elementary school [2]. When asked for their most trusted source of information, farmers named "Fellow Farmers" above all others [2]. They value information backed by "Proven Results/Experience" and "Real-world/Actual Information," indicating a preference for knowledge validated through social networks and tangible outcomes [2]. The challenge is not providing devices, but creating a digital system that can earn a place within this existing trust ecosystem.

There is clear evidence that Isabela farmers will adopt digital tools if they provide real value. A PhilRice report on a training program in remote areas of Isabela showed that farmers using agricultural mobile apps for pest and nutrient management saw significant yield increases. One farmer reported her yield rising from 1-2 tons to 2.7 tons per hectare after applying knowledge from the apps [17]. This demonstrates that digital tools can lead to productivity gains and are not automatically rejected.

However, adoption is limited by the quality of available information. Farmers report that current information is often "Too technical or difficult to understand," its truthfulness is questionable ("Unsure if true"), or there is a "Lack of information" on relevant topics [2]. This indicates a rational rejection of poorly designed or untrustworthy technology. The solution is to redesign the information and its delivery to be simple, credible, and directly applicable.

Table 2: Socio-Demographic and Technological Profile of Isabela Rice Farmers (2022)

Metric	Value
Average Age	58 years
Age Distribution (>60 years old)	38%
Gender Distribution	91% Male, 9% Female
Educational Attainment	52% High School, 24% Elementary
Average Farm Size	1.86 hectares
Smartphone Ownership	96%
Net Income (per ha/season)	15,982 pesos
Membership in Farmer Org.	31%

II. THEORETICAL FOUNDATIONS AND RELATED WORK

To help traditional farmers adopt new systems, designs should rely on proven frameworks in human behavior, technology acceptance, and inclusive design. This research combines four major theories to build a comprehensive analytical model for technology adoption.

These findings from Isabela are consistent with established models of technology acceptance:

A. The Technology Acceptance Model (TAM)

Adoption depends on Perceived Usefulness (PU) and Perceived Ease of Use (PEOU). For farmers, PU is tied to

economic benefits (higher yield, lower cost), while PEOU requires simple interfaces and local language support.

B. *The Unified Theory of Acceptance and Use of Technology (UTAUT)*

This model highlights Social Influence and Facilitating Conditions. The reliance on "Fellow Farmers" confirms the power of Social Influence. Facilitating Conditions, like phone type and signal quality, are practical constraints the system must handle.[2]

C. *The Diffusion of Innovations (DOI) Theory*

This theory classifies adopters from Innovators to Laggards. The older farmers in Isabela are likely in the Late Majority, a group that is skeptical and risk-averse. To reach them, an innovation must show a clear Relative Advantage, be Compatible with their practices, have low Complexity, be available for Trial, and have Observable results. These theories and the on-the-ground data point to a "trust and usability divide." Farmers have the devices but lack confidence in digital sources and are hindered by interfaces not designed for them. The solution is not to broadcast more data, but to build a system that digitizes the trusted dynamics of their existing peer-to-peer networks.[2]

D. *Human-Computer Interaction (HCI) Principles*

Translating adoption theories into precise design principles for the intended user group is essential. When designing interfaces for older adults, it is important to consider age-related changes in vision, motor abilities, and cognitive processes. Effective strategies include selecting large, easily readable fonts, implementing high-contrast color palettes to support visibility, providing sizable touch targets to account for diminished motor accuracy, and ensuring navigation remains straightforward and consistent to reduce cognitive demands.

Although many people have smartphones in Isabela, educational levels (52% high school, 24% elementary) indicate digital literacy is a challenge. [2] The design should be simple and intuitive, with minimal text and use of clear icons, graphics, and localized languages like Ilocano, Ybanag, Ytawis, and Yogad. The system must work well offline, storing data locally and syncing when the internet is available. To increase accessibility, an SMS-based system is vital, as it is affordable, widespread, and effective for sending alerts to those without smartphones or reliable data coverage.

Project TIWALA operates within the context of established governmental initiatives. A thorough understanding of this ecosystem is essential to clarify TIWALA's distinctive role and to pinpoint potential areas for collaboration.

1. Project SARAI (Smarter Approaches to Reinvigorate Agriculture as an Industry)

Project SARAI is a national-scale, multi-crop monitoring and forecasting program funded by the Department of Science and Technology (DOST-PCAARRD). It focuses on macro-level assessment using remote sensing and other advanced technologies to generate regional crop advisories and rainfall outlooks.

2. DA-AMIA (Adaptation and Mitigation Initiative in Agriculture)

AMIA is the Department of Agriculture's (DA) flagship program for climate change, focused on building resilience at the community level. It establishes "Climate Resilient Villages" and provides support for community-based adaptation strategies.

3. DA-BSWM (Bureau of Soils and Water Management)

The BSWM is the national authority on soil and water resources. It provides foundational data, such as official soil suitability maps and technical guidance on resource management, which are essential inputs for any localized agricultural advisory system.

A comparative analysis, as presented in Table II, identifies a distinct and strategic niche for Project TIWALA. Rather than positioning itself as a competitor to existing national programs, it serves as an essential "last-mile" application layer. The project is intended to convert regional, macro-level data from agencies such as SARAI and foundational data from BSWM into actionable, tailored decision support for individual farmers, thereby transforming national datasets into farm-level intelligence.

Table 3: Comparative Analysis of National Climate Resilience Initiatives

Initiative	Primary Objective	Synergy with TIWALA
Project SARAI	National-scale crop forecasting and monitoring for multiple crops.	Potential source of near real-time weather and crop damage data for model validation.
DA-AMIA	Community-level climate resilience building and capacity enhancement.	TIWALA can be deployed as a key technology within AMIA villages in Isabela to enhance their CRA toolkit.
DA-BSWM	Sustainable management of national soil and water resources.	Provides essential soil type and property data required for TIWALA's static risk assessment model.
Project TIWALA	Empowers farmers through easy-to-understand agricultural intelligence.	Acts as a "last-mile" application layer, translating data from other agencies into farm-level intelligence.

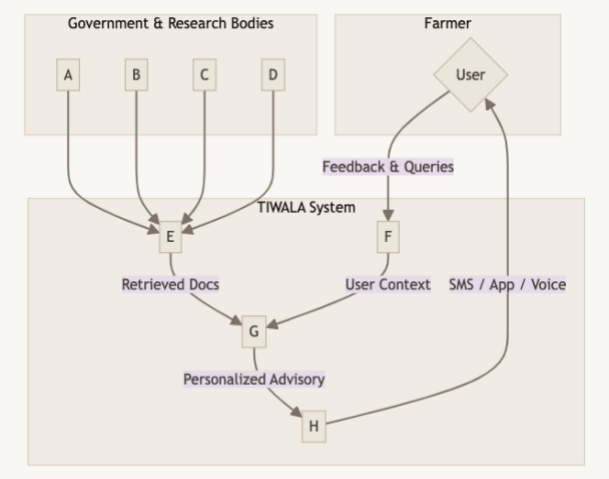
The Philippine rice sector is governed by the Rice Tariffication Law, which established RCEF to improve farmer productivity. The Department of Agriculture, along with agencies like PhilRice, PhilMech, ATI, NIA, NFA, and LBP, manages research, mechanization, training, irrigation, buffer stocks, and credit. While these agencies provide

support, information silos persist, making a unified communication platform essential.

III. RESEARCH METHODOLOGY

Project TIWALA is designed to function as an integrator within the existing ecosystem of government support programs. National initiatives like Project SARAI (crop forecasting) and DA-AMIA (community resilience), along with agencies like the Bureau of Soils and Water Management (DA-BSWM), provide valuable macro-level data that often fails to reach individual farmers in an actionable, bite-sized, simple, basic format [18]. At the provincial level, programs like the Isabela Rice Assistance Program (I-RISE) and Bojie-Rodito Opportunities (BRO) Livelihood Assistance offer direct support [19].

Figure 1: High-Level System Architecture of Project TIWALA, illustrating the flow of information from various data sources through the core AI engine to the farmer.



TIWALA is designed to fill the "last-mile" gap by ingesting, synthesizing, and translating information from these disparate sources into a coherent, personalized support system for the individual farmer.

A. *The Technical Architecture*

The core of the TIWALA system is a hyper-personalized advisory engine built on Generative AI, using a Retrieval-Augmented Generation (RAG) architecture to ensure trustworthiness and accuracy.[2]

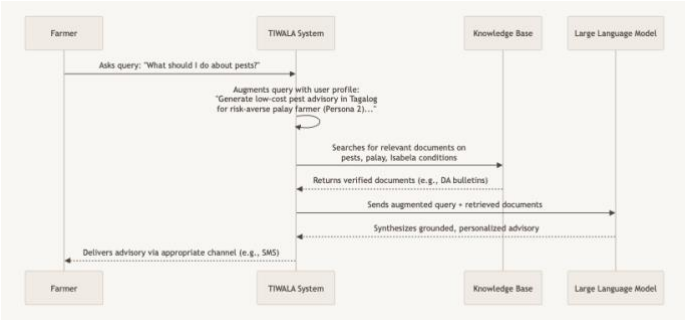
1. Data Ingestion & Knowledge Base

The system's knowledge base gathers information from authoritative sources. Web scrapers pull unstructured data like reports and advisories from the Department of Agriculture, PhilRice, and IRRI.[2] APIs provide structured, real-time data like weather forecasts from PAGASA and market prices.[2] Foundational geospatial data, such as soil fertility and pH maps for Isabela, are ingested from the DA-BSWM.[18] All information is processed and stored in a vector database, where it is segmented and converted into numerical embeddings for fast, semantic searching by the RAG system.[2]

2. Retrieval-Augmented Generation (RAG) for Trust

The RAG architecture is the system's core trust-building mechanism.[2] It prevents the AI from generating incorrect information ("hallucinating") by forcing it to base its answers on verified documents from the knowledge base. When a query is made, the system retrieves relevant documents and provides them to a Large Language Model (LLM) as context. The LLM then synthesizes an answer based only on those trusted sources.[2] This directly addresses farmers' concerns about receiving information that they are "Unsure if true".[2]

Figure 2: Workflow of the RAG-Powered Synthesis Engine, showing how a farmer's query is augmented and answered using verified information.



3. *Data-driven Persona Development using Machine Learning*

Hyper-personalization is driven by the Dynamic Farmer Profile Engine, which uses unsupervised machine learning to segment users.[2] A K-Means clustering algorithm is applied to the survey data from 49 Isabela farmers to create statistically significant personas.[2]

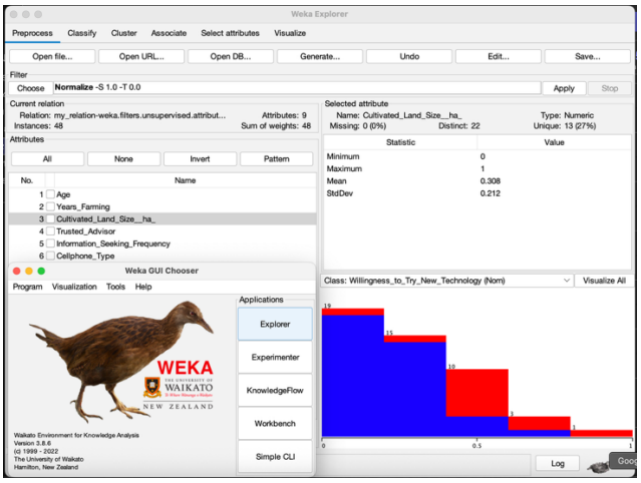
The ML pipeline includes:

- Feature Selection - Using data like age, years farming, cellphone type, and trusted information source.
- Data Preprocessing - Using One-Hot Encoding for categorical data and a StandardScaler to normalize numerical features.
- Cluster Determination - Using the Elbow Method to find the optimal number of personas.

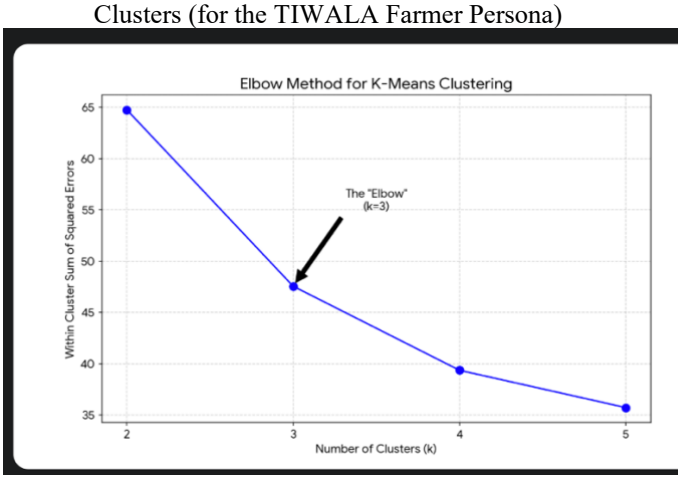
This process generates data-driven personas like "The Digital Adopter" (younger, smartphone user), "The Traditionalist" (older, basic phone user, trusts peers), and "The Community Leader" (experienced, influential).[2] These profiles are continuously updated with user interaction data, allowing recommendations to become more precise over time.[2]

Figure 3: Weka K-Means Clustering for TIWALA Farmer Personas

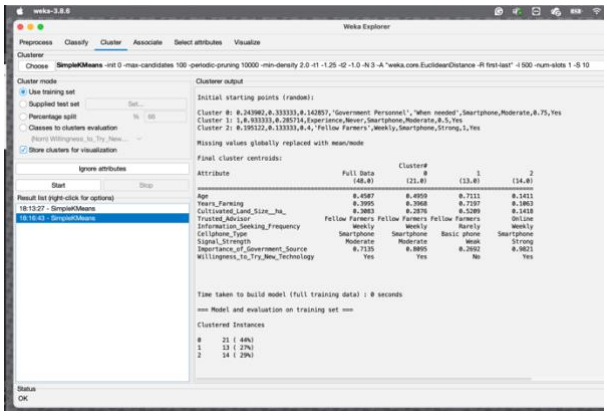
Preprocessing the Dataset of 49 Isabela Farmer Survey Respondents (Refer to Appendix A “Survey Instrument”)



"Elbow Method" to determine the optimal number of Clusters (for the TIWALA Farmer Persona)



Weka – Running the final model to get the TIWALA Farmer Persona definitions via *SimpleKMeans* algorithm with *numClusters* = **3**.



"Summary of Personas"

Attribute	TIWALA Farmer Persona 1 "Traditionalist Veteran"	TIWALA Farmer Persona 2 "The Digital Adopter"	TIWALA Farmer Persona 3 "The Pragmatic Majority"
Cluster	Cluster 1 (27%)	Cluster 2 (29%)	Cluster 0 (44%)
Key Trait	Skeptical, Experienced, Rich	Tech-Savvy, Novice, Small	App-Ready, Peer-Influenced, Average
Phone	**Basic Phone**	Smartphone	Smartphone
Trusts	(No one listed)	**Online** / Government	**Fellow Farmers** / Government
Tech?	**No**	**Yes**	**Yes**
Gov. Trust	**Lowest**	**Highest**	High
TIWALA Role	***SMS-Only, Laggard***	***App Innovator, Pilot User***	***App Early Majority, Key to Scale***

4. Multi-Channel Delivery

The system delivers advisories through the channel best suited to each farmer's profile. A "Traditionalist" with a basic phone might receive a simple SMS, while a "Digital Adopter" receives a richer notification via the mobile app. This ensures information is accessible to all users, regardless of their digital literacy or device limitations.

5. VALIDATION, OUTCOMES, AND STRATEGIC PATH FORWARD

The TIWALA framework will be validated through a pilot evaluation with 200 farmers in Isabela, testing both the mobile app and SMS-only channels.

- Usability (Ease of Use)

Measured by Task Success Rates (target: ⩾70% for app, ⩾45% for SMS) and Task Completion Time (target: under 60 seconds to get a recommendation). [31]

- Effectiveness (Usefulness)

The primary KPI is a target of at least 40% of users reporting a changed farming decision based on a TIWALA advisory. Case studies will document specific instances where advisories led to positive economic outcomes, such as avoiding pest damage or securing better prices. [31]

The project aligns with the strategic goals of both provincial and national governments. The Provincial Government of Isabela's "ROAD TO AGRI-INNOVATION" vision calls for the adoption of "advanced technologies" and "science-based and climate-resilient farming technologies".[19] At the national level, the Department of Agriculture's "One DA" agenda promotes modernization through digital agriculture and improved supply chains.[26] TIWALA directly supports these mandates by providing a tangible tool for modernizing the agricultural information supply chain.

To move from concept to implementation, strategic partnerships are essential:

1. Formal Partnership with the Department of Agriculture (DA) - To gain official endorsement, facilitate access to farming communities, and ensure access to authoritative data streams.
2. Integration with the DA-AMIA Program - To deploy TIWALA as a key technology within the Climate Resilient Villages in Isabela, embedding it in an existing community-focused framework.[28]
3. Data-Sharing with Project SARAI - To leverage SARAI's macro-level climate data to improve the accuracy of TIWALA's farm-level predictive models.
4. Alliance with the Provincial Government of Isabela (PGI): To use the TIWALA platform as an official channel for disseminating information about provincial programs like I-RISE and BRO, gaining local credibility in return.[19]

Future work will focus on a year-long study to assess the system's impact on full harvest cycles and farmer profitability. The long-term goal is to expand the system from reactive advice to proactive, long-term planning support, offering guidance on crop diversification and sustainable water management to build lasting climate resilience.[31]

6. CONCLUSION AND FUTURE WORK

This paper presents Project TIWALA, a socio-technical framework that will use practical Generative AI, RAG, and machine learning to make agricultural intelligence more accessible for smallholder rice farmers in Isabela, Philippines. This research is expected to deliver significant, tangible impacts across multiple sectors. For farmers, it provides a direct pathway to reduce crop losses, protect investments, and stabilize incomes, thereby enhancing livelihood and resilience. For governance, it offers a data-driven tool for local government to improve agricultural planning, issue targeted advisories, and strategically allocate resources. For science, it contributes a novel, integrated framework for agricultural risk assessment and a unique, high-resolution dataset for regional climate research.

The TIWALA framework, with its foundational emphasis on building trust and its rigorous user-centered design methodology, offers a potentially replicable model for technology adoption in other agricultural contexts, both within the Philippines and in other developing regions. Many smallholder farming communities globally face similar socio-technical challenges: a demographic of older, traditional farmers, limited digital literacy, poor connectivity, and a deep-seated skepticism towards new technologies. It provides a robust, empathetic, and safe model for bridging the "last-mile" information gap—a foundational step toward creating a more informed, resilient, and empowered global farming community.

The main limitation of this conceptual framework is that its effectiveness can only be determined after the planned pilot study is carried out. The geographic focus on rice farmers in Isabela and the short duration of the proposed pilot may limit the generalizability of the findings to other regions, crop types, or across different cropping seasons with varying climatic conditions. The pilot relies on self-reported behavioral changes and system logging.

Future work will address current limitations and enhance the framework. A year-long study is planned to assess its impact on harvests, long-term adoption, and farmer profitability. An important goal is to broaden the framework so it covers proactive climate adaptation strategies. This means shifting focus from short-term advice to comprehensive long-term planning, offering guidance on areas like crop diversification and sustainable water management. There is also significant potential to incorporate more advanced machine learning for predictive analytics tailored to local conditions. This could involve developing models that predict pest and disease outbreaks by combining satellite data, ground-level sensor data from networks like PhilSensors, and user-reported observations from the "Bayanihan" community feature.

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REFERENCES

[1] Official Website of the Province of Isabela - Economic Profile, accessed September 15, 2025, https://provinceofisabela.ph/index.php/doing-business-in-isabela/2013-07-09-01-13-03 [2] Clean Survey Data - Final - Translated.csv [3] Philippine Statistics Authority | Republic of the Philippines, accessed October 1, 2025, https://psa.gov.ph/statistics/vop-main/provincial/node/1684065243 [4] Provincial Value of Production in Agriculture and Fisheries, 2024 ..., accessed September 28, 2025, https://psa.gov.ph/content/provincial-value-production-agriculture-and-fisheries-2024 [5] Philippine Statistics Authority | Republic of the ... - Psa.gov.ph, accessed September 10, 2025, https://psa.gov.ph/statistics/ppa/node/1684061782 [6] A Review of the Agriculture Sector in Cagayan Valley | Philippine ..., accessed October 5, 2025, https://psa.gov.ph/content/review-agriculture-sector-cagayan-valley [7] Rice Industry - Ricelytics, accessed September 19, 2025, https://ricelytics.philrice.gov.ph/rice_industry [8] Rice - Industry Strategic Science and Technology Plans (ISPs) Platform, accessed September 22, 2025, https://ispweb.pcaarrd.dost.gov.ph/isp-commodities/rice/ [9] Industry Analysis: Agriculture in Philippines - Asia School of Business (ASB), accessed October 3, 2025, https://asb.edu.my/wp-content/uploads/2025/02/BCS010-Agriculture-in-the-Phillippines.pdf [10] Cagayan Valley's 2024 Palay Production Drops by 3.4 Percent, accessed September 12, 2025, https://rsso02.psa.gov.ph/content/cagayan-valleys-2024-palay-production-drops-34-percent [11] (PDF) Climate Risk Vulnerability Assessment of the Agriculture ..., accessed September 30, 2025, https://www.researchgate.net/publication/363796616_Climate_Risk_Vulnerability_Assessment_of_the_Agriculture_Sector_in_Isabela_Province_Basis_for_Decision_Support_Platform_and_Policy_Innovation [12] Building a Dam and Irrigation System to Help Farmers in Isabela, the Philippines, Adapt to Climate Change, 2011–18 - Global Partnership for Effective Development Cooperation, accessed September 8, 2025, https://www.effectivecooperation.org/content/building-dam-and-irrigation-system-help-farmers-isabela-philippines-adapt-climate-change [13] Building a dam and irrigation system to help farmers in Isabela, the Philippines, adapt to climate change, 2011–18 - K-Developedia, accessed October 6, 2025, https://www.kdevelopedia.org/Resources/view/99202006300152918.do [14] Climate change impacts on spatiotemporal variation of extreme weather and its consequences on dam optimisation and risk management in the Cagayan basin, Philippines, accessed September 14, 2025, https://www.apn-gcr.org/publication/climate-change-impacts-on-spatiotemporal-variation-of-extreme-weather-and-its-consequences-on-dam-optimisation-and-risk-management-in-the-cagayan-basin-philippines/ [15] Climate change impacts on spatiotemporal variation of extreme weather and its consequences on dam optimisation and risk management in the Cagayan basin, Philippines, accessed September 17, 2025, https://www.apn-gcr.org/bulletin/article/climate-change-impacts-on-spatiotemporal-variation-of-extreme-weather-and-its-consequences-on-dam-optimisation-and-risk-management-in-the-cagayan-basin-philippines/ [16] NIA switches on P1.029-billion Pasa SRIP | National Irrigation ..., accessed September 25, 2025, https://www.nia.gov.ph/content/nia-switches-p1029-billion-pasa-srip [17] Farmers in Isabela, Abra reap gains from agri-apps : DA-PhilRice, accessed October 2, 2025, [suspicious link removed] [18] Maps - BSWM, accessed September 11, 2025, http://www.bswm.da.gov.ph/maps/ [19] ROAD TO AGRI-INNOVATION - Province of Isabela, accessed September 20, 2025, https://provinceofisabela.ph/images/INFOcus/INFOcus_PDF/INFOCUS%202024%20FINAL.pdf [20] Isabela Recovery Initiatives to Support Enterprises (I-RISE), accessed October 7, 2025, https://provinceofisabela.gov.ph/flagship-programs-and-projects/isabela-recovery-initiatives-to-support-enterprises-i-rise/ [21] Official Website of the Province of Isabela - BOJIE-RODITO OPPORTUNITIES, accessed September 29, 2025, https://provinceofisabela.ph/index.php/using-joomla/extensions/components/content-component/article-categories/310-bropportunities?showall=1&limitstart= [22] Government program helps Isabela farmers avail of interest-free loan - Agriculture Monthly, accessed September 16, 2025, https://agriculture.com.ph/2019/04/14/government-program-helps-isabela-farmers-avail-of-interest-free-loan/ [23] BSWM MAPS - Department of Agriculture, accessed September 24, 2025, https://www.bswm.da.gov.ph/bswm-maps/ [24] ISABELA_soil-ph-rice_201709 - BSWM - Department of Agriculture, accessed October 4, 2025, https://www.bswm.da.gov.ph/map/isabela-soil-ph-rice-201709/ [25] ISABELA_soil-ph-corn_201909 - BSWM - Department of Agriculture, accessed September 13, 2025, https://www.bswm.da.gov.ph/map/isabela-soil-ph-corn-201909/ [26] (PDF) Transforming Philippine agriculture through Agriculture 4.0 - ResearchGate, accessed September 27, 2025, https://www.researchgate.net/publication/377351004_Transforming_Philippine_agriculture_through_Agriculture_40 [27] Inspiring Agribusiness Stories - Department of Agriculture, accessed September 9, 2025, https://www.da.gov.ph/wp-content/uploads/2022/04/AMAS-Inspiring-Agribusiness-Stories-04202022.pdf [28] Scaling the capacities to adapt to a changing climate | IIRR, accessed September 18, 2025, https://iirr.org/wp-content/uploads/2021/09/Info-Note_Scaling-the-capacities-to-adapt-to-a-changing-climate-Experiences-of-the-AMIA-Climate-Resilient-Villages-Philippines.pdf [29] DA RFO 02 capacitates AMIA Villagers for a climate-resilient rice production, accessed October 6, 2025, https://cagayanvalley.da.gov.ph/2021/10/14/da-rfo-02-capacitates-amia-villagers-for-a-climate-resilient-rice-production/ [30] Digital agriculture for climate resiliency - DA-PhilRice, accessed September 23, 2025, https://www.philrice.gov.ph/digital-agriculture-for-climate-resiliency/ [31] P. Gumpal, "Project TIWALA: A Socio-Technical Framework for Democratizing Agricultural Intelligence in Isabela," unpublished, 2025.

APPENDIX A: RESEARCH INSTRUMENTS AND PROTOTYPES

This appendix contains the survey questionnaire distributed to farmers and a selection of high-fidelity prototype screens for the TIWALA application.

A.1 Original Survey Instrument

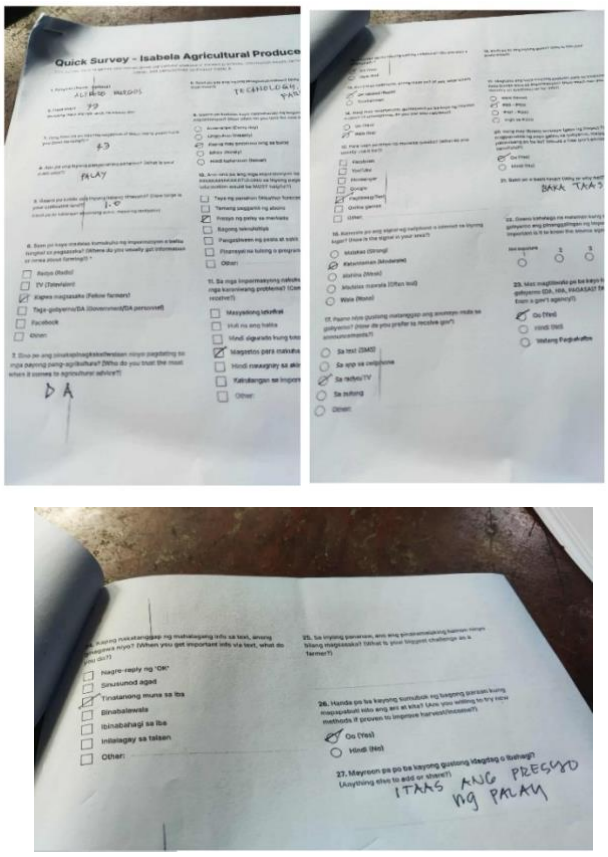

Figure 5: The original survey form distributed to farmers.

A.2. High-Fidelity Prototype Screens

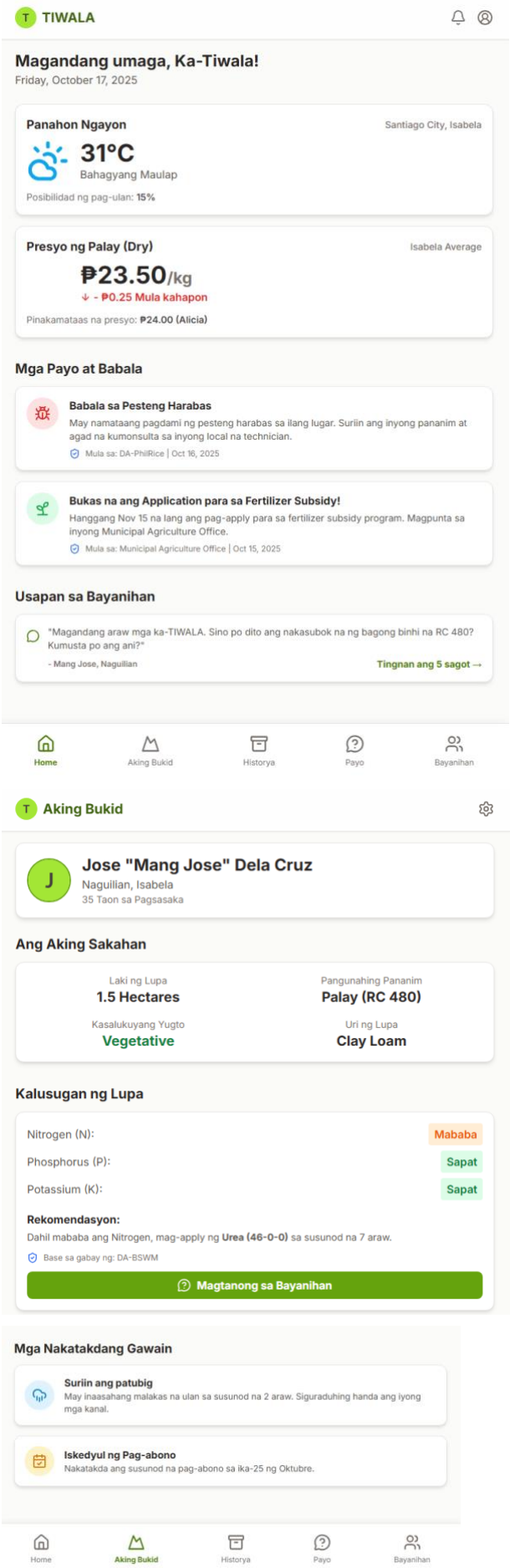

Figure 6: The Main Dashboard screen. This design prioritizes Perceived Usefulness (PU) by displaying the most requested information—weather and market prices—at the forefront. The clean layout with large icons addresses Perceived Ease of Use (PEOU).

Figure 7: The Historical Data ("Historya") screen. This screen builds Trust by providing transparent access to past advisories and seasonal outcomes, allowing farmers to learn from historical trends and verify the system's track record over time.

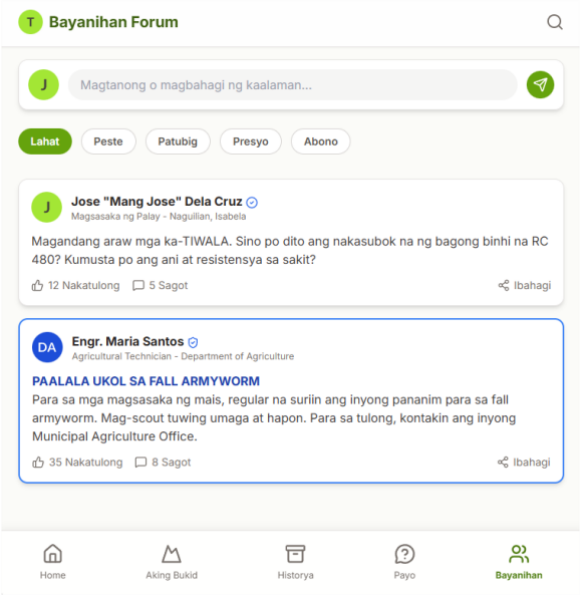

Figure 8: The Community Network ("Bayanihan") screen. This feature is designed to leverage Social Influence. By creating a professional network for verified farmers, it digitizes the trusted peer-to-peer knowledge sharing identified in the survey and provides a safe, productive alternative to open social media platforms.